

Mobile Forensics

Module V

Introduction of Mobile Forensics

Process of extracting and analyzing information, which had been stored in mobile devices.

Information obtained via mobile device forensics may include deleted files, application data, GPS data, call logs, text messages, and images and videos.

Commonly used to recover evidence in connection with a criminal investigation mobile device forensic investigators must take care to retrieve and analyze data that is legally admissible as evidence.

mobile device forensic analysts must be intimately familiar with mobile devices and their operating systems and file systems.

They should also have experience with various software and hardware tools for extracting data from mobile devices

Process of Mobile Device Forensics

1. Device seizure
2. Device acquisition
3. Device analysis
4. Reporting

1. Device seizure

the mobile device is seized from its user.

At this stage, investigators should also start documenting the chain of custody.

For example, the records of who handled the device and when.

A search warrant is usually required if the device is used in a criminal investigation.

2. Device acquisition

Investigators create a sector-level duplicate of the device, a process known as “imaging” or “acquisition.”

This duplicate image and the original device are passed through a hashing function, and their outputs are compared to ensure that it is an exact copy.

Next, analysts decide on the investigation’s proper approach and goals.

Types of Mobile Cloning in Forensics

Logical Acquisition – Copy visible data (contacts, SMS, call logs, app data)

File System Acquisition – Copy full file system

Physical Acquisition – Bit-by-bit clone of entire memory

Manual Acquisition – Screenshot/documentation method

3. Device analysis

Investigators begin work on the device image to confirm a hypothesis or search for hidden data.

Specialized tools (such as those described in the next section) are used to help find and recover information.

Data may be located within the accessible hard disk space, deleted (unallocated) disk space, or the operating system cache.

4. Reporting

The document or paper trail that shows the seizure, custody, control, transfer, analysis, and disposition of physical and electronic evidence is referred to as forensic reporting.

It is the process of verifying how any type of evidence was collected, tracked, and protected.

Analysis of Evidences

Specialised forensic discipline within digital investigations that focuses on extracting and examining electronic evidence from mobile communication devices.

Encompass a comprehensive approach to identifying, acquiring, processing, and reporting electronically stored information for legal proceedings.

The core objectives of mobile device forensics include:

- Recovering deleted or hidden digital evidence
- Preserving data integrity during extraction
- Analysing communication records and metadata
- Documenting forensic processes for potential court presentation
- Decrypting secured device information

Source of Evidence

Device Hardware (Physical Evidence)

Examples:

Mobile handset (internal memory)

SIM card

SD / microSD card

Battery (may store logs in some cases)

Evidence Found:

Device identifiers (IMEI, serial number)

Stored data in flash memory

Deleted data (recoverable)

SIM contacts and SMS

SIM Card

Data Available:

Contacts

SMS messages

IMSI number

Last dialed numbers

Service provider details

Note: Even if the phone is damaged, SIM may still contain useful information.

Internal Storage (Flash Memory)

This is the main source of evidence.

Call logs

SMS / MMS

Photos & videos

WhatsApp chats

App data

Browser history

Downloaded files

Deleted files (recoverable)

External Storage (Memory Card)

Sometimes criminals store data in hidden folders.

Images

Audio recordings

Videos

Documents

Backup files

App media folders

Installed Applications

Common Apps Investigated:

Messaging apps (WhatsApp, Telegram)

Social media apps

Banking apps

Email apps

Cloud storage apps

Evidence Found:

Chat history

Shared media

Login credentials

Transaction history

Network & Communication Data

Wi-Fi connection logs

Bluetooth pairing history

GPS location data

IP address logs

Browsing history

This helps track:

Location of suspect

Communication patterns

Internet usage

Cloud Storage & Backups

Cloud data is often requested legally from service providers.

Even if data is deleted from the phone, it may exist in:

Google Drive

iCloud

Dropbox

OneDrive

Service Provider Records

Collected from:

Telecom operators

Internet Service Providers

Includes:

Call Detail Records (CDR)

SMS logs

Tower location data

Subscriber details

Logs and System Files

System logs

Error logs

App usage logs

Authentication logs

These help determine:

When the phone was used

Which app was accessed

Login attempts

Challenge in Mobile Forensics

The biggest challenge in mobile forensics is keeping up with the rapid pace of change in mobile technology.

New devices and operating systems are constantly being released, each with its unique file system and data storage methods.

This makes it challenging for mobile forensics experts to stay current with the latest changes.

Differences in hardware: Mobile devices come in all shapes and sizes, with different types of hardware. This can make it difficult to develop mobile forensics tools that work on all devices.

Password security and encryption: Many mobile devices are password-protected and encrypted, making data recovery and mobile forensics difficult.

Mobile operating systems : There are many different mobile operating systems, each with its own file system and data storage methods. This can make data acquisition and interpretation difficult.

Accidental device reset: One of the most common problems mobile forensics experts face is when a user accidentally resets their device. This can delete all the device files and log files, making it difficult to recover key evidence.

Lack of tools and equipment: Mobile forensics is still a relatively new field, and relatively few tools exist. This makes it difficult to perform mobile forensics efficiently.

Anti-forensic techniques: As mobile forensics becomes more popular, criminals are also becoming more aware of it and are using anti-forensic techniques to prevent their data from being recovered.

Mobile platform security features: Many mobile devices have built-in security features that can make data recovery difficult. For example, Apple's iPhone has a "Secure Enclave" feature that encrypts all the data on the device.

Preventing data modification: One goal of mobile forensics is to preserve data on a mobile device so it can be used as evidence in court. However, this can be difficult if the data is constantly modified. Many mobile devices automatically delete old data to make room for new data, making it difficult to recover deleted data.

Legal issues: Mobile device forensics can be used to recover sensitive electronic evidence for use in the legal process. However, many laws govern how this data can be used. For instance, there are laws that protect a user's privacy.

The dynamic nature of evidence: Mobile devices constantly change, making it difficult to keep track of all the data on them. A user might install a new app or delete an old one, which can change the data on the device. This makes it difficult to know what data is relevant and what isn't.

Device alteration: Mobile devices can be easily altered. For example, a user might root their device, which modifies the default data on the device and makes it difficult to recover.

Communication shielding: This is when a user uses a mobile device to communicate with someone they don't want to be tracked. For example, they might use a burner phone or an encrypted messaging app. This can make it difficult to recover the data from the mobile device.

Malicious programs: These programs are designed to prevent mobile forensics experts from accessing data on a mobile device. For example, a user might install a program that encrypts all the data on their device.

Tools used in Mobile Forensics

1. Commercial / Professional Mobile Forensic Tools

These are full-featured suites commonly used in criminal, civil, and corporate investigations:

Cellebrite UFED: Industry-standard extraction and analysis platform that can perform *logical and physical data extractions* from a wide range of mobile devices, including password/passcode bypass under legal authority.

MSAB XRY: Widely used to acquire and decode data from smartphones and tablets (iOS, Android, feature phones). Offers *logical, file-system, and physical data extraction* with strong chain-of-custody support.

Magnet AXIOM: A digital forensics platform that integrates *mobile and computer evidence*. AXIOM pulls artifacts from phones, cloud services, and computers and correlates them for easier interpretation.

Oxygen Forensic Detective: Comprehensive analysis suite that extracts data from devices and cloud backups, with timeline construction and social graphing features.

MOBILedit Forensic: Tool capable of extracting detailed phone data — SMS, call logs, contacts, multimedia, and device identifiers — from many phone brands and OSes.

Belkasoft Evidence Center X: A DFIR (Digital Forensics and Incident Response) software that not only extracts mobile device data but also analyzes timelines and inter-device connections.

Elcomsoft Mobile Forensic Bundle: Includes tools specialized for deep iOS and Android extraction and decryption. (Listed in industry tool reviews)

Paraben E3: Forensic suite for acquisition and analysis across devices, including mobile phones.

2. Supplementary & Specialized Tools

These tools are often used in specific workflows or to support main suites:

Autopsy (with mobile modules): Open-source digital forensics platform; mobile modules can parse Android or iOS data dumps, useful for analysis and verification.

ADB (Android Debug Bridge): Command-line tool for Android device interaction that can help extract data when combined with scripts or analysis pipelines.

AFLogical OSE: Open-source tool for Android logical data extraction, often used in resource-constrained or education settings.

Project A Phone / ZRT2 / Eclipse: Older but useful extraction utilities referenced in mobile forensics literature.

3. Supporting Utilities & Techniques

These aren't full suites but support analysis and extraction:

- **SQLite Browser (DB Browser for SQLite):** Handy for inspecting databases extracted from devices.
- **iLEAPP / ALEAPP:** Open-source tools for processing iOS and Android artifact data (community-referenced).
- **Bulk Extractor:** Extracts artifacts (emails, URLs, phone numbers) from binary images and dumps.
- **Linux tools (e.g., Autopsy, Sleuth Kit):** Useful in forensic analysis workflows.

4. Types of Mobile Forensic Tool Functions

Understanding what each tool does helps you choose the right one.

- **Data Acquisition:** Extracts data from device memory — logical, file system, or physical.
- **Data Analysis:** Parses and interprets extracted data (messages, call logs, app artifacts).
- **Reporting:** Generates forensic reports acceptable in legal cases.